Frequentist and Bayesian Reanalysis of the Pfizer BNT162b2 Vaccine Trial

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03/14/2026

Abstract

COVID-19 caused major global health problems and created an urgent need for effective vaccines. In this study, we reviewed the Pfizer–BioNTech BNT162b2 vaccine trial reported by Polack et al. (2020) and reanalyze its efficacy using both frequentist and Bayesian methods. Using the observed data of 8 infections in the vaccine group and 162 in the placebo group, we estimate the vaccine efficacy parameter and evaluate whether it exceeds the FDA benchmark of 30%. The frequentist analysis uses maximum likelihood estimation, a large-sample confidence interval, and a likelihood ratio test. The Bayesian analysis uses a beta-binomial model with prior elicitation, posterior summaries, and a posterior probability calculation. Both approaches show strong evidence that the vaccine efficacy is substantially greater than 30%. Our findings confirm the high effectiveness of the BNT162b2 vaccine but call into question if BNT162b2's efficacy was overestimated based on Pfizer's chosen prior. We recommend more rigorous and transparent discussion of priors in future studies that apply Bayesian methods.

Keywords

Bayesian Inference, *Likelihood Inference*, *COVID-19*, *Vaccine efficacy*

Introduction

Beginning in 2019, the COVID-19 pandemic has had a profound impact on global public health. It has been estimated that approximately 18.2 million people died worldwide as a result of the pandemic (Wang, Haidong et al. 2020-21). Vaccination therefore became a critical strategy for reducing infection and mortality. There are studies showing that "Authorized mRNA COVID-19 vaccines are effective for preventing SARS-CoV-2 infection in real-world conditions." and COVID-19 vaccination is recommended for all eligible persons." (Thompson et al., 2021). One of the most widely used vaccines is the BNT162b2 mRNA vaccine developed by Pfizer and BioNTech.

Vaccine efficacy measures the proportional reduction in infection risk among vaccinated individuals relative to those receiving a placebo. The U.S. Food and Drug Administration (FDA) recommended that COVID-19 vaccines demonstrate at least 30% efficacy to be considered effective. The efficacy of this vaccine was evaluated in a large randomized, placebo-controlled, double-blind clinical trial (Polack et al., 2020), which randomly assigned participants aged 16 years or older in a 1:1 ratio to receive two doses, 21 days apart, of either placebo or the BNT162b2 vaccine. The primary efficacy endpoint was the occurrence of laboratory-confirmed COVID-19 infection. A total of 170 infections were targeted (Polack et al., 2020).

Transparent, corroborated methodology is vital to global perceptions of public health institutions. The mixed messaging during the pandemic led to a deterioration in trust of public health (Ortiz-Prado et al., 2025). In our report, we aim to corroborate and validate Pfizer's analysis using both Frequentist and Bayesian statistical approaches. The Pfizer vaccine trial was notable in its use of Bayesian methods, whereas other firms used frequentist experimental design. Pfizer's study has been criticized for lacking explanation of study assumptions (most notably the choice of a prior) and pandemic vaccine trials have been criticized for a lack of data transparency (Senn S., 2022). To provide an independent statistical assessment of Pfizer's BNT162b2 vaccine efficacy, we estimate the vaccine efficacy parameter and test whether the efficacy exceeds the FDA guideline of 30%.

Table 1: vaccine efficacy at least 7 days after dose two

Group	cases	N	rate
BNT162b2	8	17411	0.00046
Placebo	162	17511	0.00925

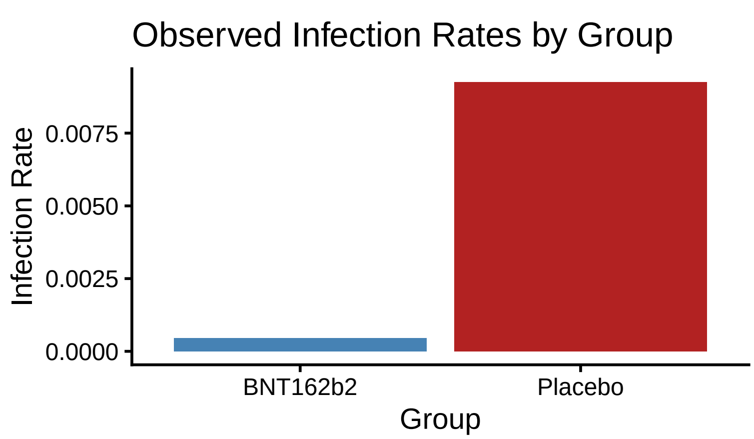


Figure 1: Observed Infection Rates by Group

Statistical Methods

Model

We define V as the number of people who were infected with COVID-19 that received the vaccine. n is the total number of test subjects that were infected (170 in our case) and π_0 is the true but unknown probability of being vaccinated given that they are infected.

This true probability has been calculated in terms of two other probabilities, π_{v0} and π_{p0}, which are the proportions of COVID infections, for the vaccine and placebo groups respectively, and since the trial is approximately 1:1 randomized, we have

$$\pi_0 = \frac{\pi_{v0}}{\pi_{v0} + \pi_{p0}}$$

What we are interested in is the efficacy of the vaccine, ψ_0. The estimate of this efficacy, ψ, is computed as the inverse of the estimated relative risk. That is:

$$\psi = 1 - \frac{\pi_v}{\pi_p} = \frac{1 - 2\pi}{1 - \pi}$$

We can use this to transform π in terms of ψ (which is more helpful for our inferences):

$$\pi = \frac{1 - \psi}{2 - \psi}$$

Likelihood Inference

Likelihood function and MLE

In this section, we take the likelihood approach and break down the problem through a frequentist's lens. We begin by approximating V, the number of subjects infected with COVID-19 that were vaccinated, to the binomial distribution (proof in appendix):

$$V \approx Binom(n, \pi_0)$$

From Table 1, we know $n = 170$ and observe $v = 8$. From lecture, we know the MLE of a binomial distribution ratio of successes to trials (Grove, 2026, Lecture 11). Hence,

$$\widehat{\pi}_0^{mle} = \frac{8}{170} \quad \widehat{\psi}_0^{mle} = \frac{1 - 2(\frac{8}{170})}{1 - (\frac{8}{170})} \approx 0.9506$$

We can verify this with the transformed likelihood function, $L^*(\psi)$. We begin with the binomial likelihood function, and transform it to be in terms of ψ:

$$
\begin{aligned}
L(\pi) &= \binom{n}{v}\pi^v(1-\pi)^{n-v} \\
L^*(\psi) &= \binom{n}{v}(\frac{1-\psi}{2-\psi})^v(1-\frac{1-\psi}{2-\psi})^{n-v} \\
L^*(\psi) &= \binom{n}{v}(\frac{1-\psi}{2-\psi})^v(\frac{1}{2-\psi})^{n-v} \\
L^*(\psi) &= \binom{n}{v}\frac{(1-\psi)^v}{(2-\psi)^n} \\
\ell^*(\psi) &= \ln(\binom{n}{v}) + v\ln(1-\psi) - n\ln(2-\psi) \\
\frac{d}{d\psi}\ell^*(\psi) &= -\frac{v}{1-\psi} + \frac{n}{2-\psi}
\end{aligned}
$$

The above gives us the log-likelihood function $\ell^*(\psi)$ and its derivative. We find a critical point by equating the derivative to zero, but we would still need to verify it to be a maximum (and thus, our MLE).

Second Derivative test

This is where the second derivative test comes in. We check the value of the second derivative at the critical point to identify the concavity, and whether or not we found a maximum:

$$\frac{d^2}{d\psi^2}\ell^*(\psi) = -\frac{v}{(1-\psi)^2} + \frac{n}{(2-\psi)^2}$$

If the above second derivative is less than 0 at our critical point (or for all values of ψ), then it is concave down and we have verified it to be our MLE, $\widehat{\psi}_0^{mle}$

Large Sample Confidence Interval

The next step of our likelihood inference is finding the confidence interval, for which we take the large sample approach. Before we do so, we plot the second-order Taylor approximation against the log-likelihood of π to ensure our approximations hold:

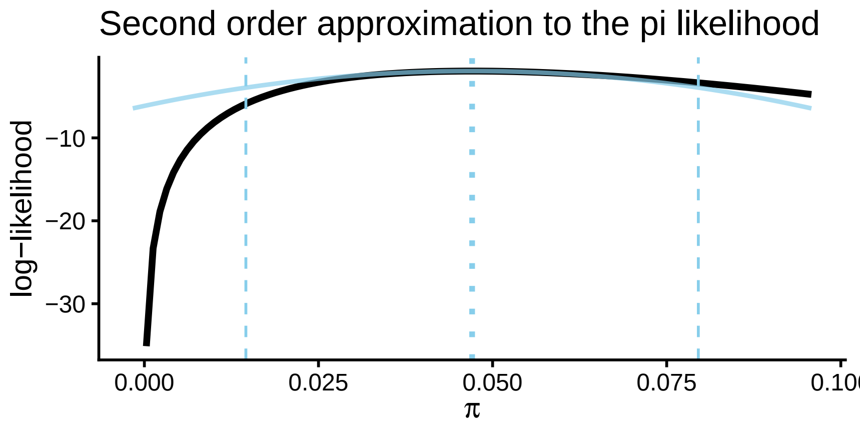


Figure 2: Log−likelihood function and second−order approximation for pi.

We see the second-order Taylor approximation is accurate, mostly aligned with the log-likelihood. Thus, we move forward with the large sample confidence interval. To do so, we need to calculate our observed Fisher Information:

$$I(\psi) = -E\left[\frac{d^2}{d\psi^2}\ell^*(\psi)\right]$$

$$I(\psi) = E\left[\frac{v}{(1-\psi)^2} - \frac{n}{(2-\psi)^2}\right]$$

$$I(\psi) = \frac{E[v]}{(1-\psi)^2} - \frac{n}{(2-\psi)^2}$$

Note that: $E[v] \approx n * \pi$

$$I(\psi) = \frac{n*\pi}{(1-\psi)^2} - \frac{n}{(2-\psi)^2}$$

$$I_{obs}(\widehat{\psi}_0^{mle}) = \frac{(n - n\widehat{\psi}_0^{mle})(2-\widehat{\psi}_0^{mle}) - n(1-\widehat{\psi}_0^{mle})^2)}{(1-\widehat{\psi}_0^{mle})^2(2-\widehat{\psi}_0^{mle})^2}$$

We then compute the 95% large sample confidence interval with the formula:

$$\widehat{\psi}_0^{mle} \pm z_{\alpha/2}\sqrt{\frac{1}{I(\widehat{\psi}_0^{mle})}}$$

Significance test: Likelihood Ratio Statistic

For the significance test, we want to find the likelihood ratio statistic W to test whether or not our efficacy rate passes the FDA's standard. We define our hypotheses as follows:

$$H_0 : \psi_0 = 0.3 \quad H_1 : \psi_0 \neq 0.3$$

We know $W = 2\ln(\Lambda)$, where $\Lambda = \frac{L(\widehat{\psi}_0^{mle})}{L(\psi_0^{null})}$

$$
\begin{aligned}
W &= 2\ln\left(\frac{L(\widehat{\psi}_0^{mle})}{L(\psi_0^{null})}\right) \\
W &= 2\ln\left(\frac{\binom{n}{v}\frac{(1-\widehat{\psi}_0^{mle})^v}{(2-\widehat{\psi}_0^{mle})^n}}{\binom{n}{v}\frac{(1-\psi_0^{null})^v}{(2-\psi_0^{null})^n}}\right) \\
W &= 2\ln\left(\frac{(1-\widehat{\psi}_0^{mle})^v(2-\psi_0^{null})^n}{(2-\widehat{\psi}_0^{mle})^n(1-\psi_0^{null})^v}\right) \\
W &= 2(v\ln(1-\widehat{\psi}_0^{mle}) + n\ln(2-\psi_0^{null}) - n\ln(2-\widehat{\psi}_0^{mle}) - v\ln(1-\psi_0^{null}))
\end{aligned}
$$

We then take the observed w and compare it against the chi-squared distribution of W. That is, our p-value will be $P(W \geq w)$, where $W \sim \chi_1^2$. At the same time, we conduct a parametric bootstrap to find a bootstrapped confidence interval and empirical p-value, for a more robust analaysis into our MLE.

This concludes our the methods of our likelihood inference approach, and the results of these procedures will be presented under the Results section.

Bayesian Inference

In this section we describe the Bayesian approach used to estimate the vaccine efficacy parameter ψ, where $-\infty < \psi \leq 1$.

To complete the Bayesian model, we specify a Beta prior for Π:

$$\Pi \sim \text{Beta}(a, b).$$

The Beta family is a natural prior for a binomial parameter because it is supported on $[0, 1]$ and is conjugate to the binomial likelihood.

We chose the prior through elicitation on the vaccine efficacy parameter ψ rather than directly on π. Our goal was to use a skeptical prior that would require the observed data to provide strong evidence before concluding that the vaccine was effective. This choice is reasonable because, at the design stage of a clinical trial, substantial uncertainty remains about whether a candidate vaccine will exceed the FDA efficacy benchmark of 30%.

Specifically, we imposed the following prior beliefs:

1. $P(\Psi \geq 0.3) = 0.1$,
2. $P(\Psi \geq 0) = 0.5$.

The first statement assigns only a 10% prior probability to the event that vaccine efficacy exceeds the FDA benchmark, which shows skepticism about strong efficacy before seeing the trial data. The second statement centers prior belief at the boundary between harmful and beneficial effects by assigning probability 0.5 to $\Psi \geq 0$.

Using the transformation

$$\psi = \frac{1-2\pi}{1-\pi} \quad \Longleftrightarrow \quad \pi = \frac{\psi-1}{\psi-2} \qquad (1)$$

these prior beliefs become

$$P\left(\Pi \leq \frac{7}{17}\right) = 0.1, \qquad P\left(\Pi \leq \frac{1}{2}\right) = 0.5.$$

Solving these equations numerically gives

$$\Pi \sim \text{Beta}(26.2, 26.2).$$

Because the Beta prior is conjugate to the Binomial likelihood, by theorem 13.2 (Grove, 2026, Lecture 13), the posterior distribution of Π is

$$\Pi \mid V = v \sim \text{Beta}(26.2 + 8,\ 26.2 + 162) = \text{Beta}(34.2, 188.2).$$

To obtain the posterior distribution of Ψ, we use the transformation (1)

For $-\infty < \psi \leq 1$, the cumulative distribution function of Ψ is

$$\begin{aligned} P(\Psi \leq \psi | V = v) &= P(\frac{1-2\Pi}{1-\Pi} \leq \psi | V = v) \quad -\infty < \psi \leq 1 \\ &= 1 - P(\Pi \leq \frac{\psi-1}{\psi-2} | V = v) \\ F_{\Psi|V}(\psi|v) &= 1 - F_{\Pi|V}(\frac{\psi-1}{\psi-2}|v) \end{aligned}$$

Differentiating gives the posterior density of ψ:

$$f_{\Psi|V}(\psi|v) = (1 - F_{\Pi|V}(\frac{\psi-1}{\psi-2}|v))'(\frac{\psi-1}{\psi-2})'$$
$$f_{\Psi|V}(\psi|v) = f_{\Pi|V}(\frac{\psi-1}{\psi-2}|v)\frac{1}{(\psi-2)^2} = \frac{(1-\psi)^{a-1}}{B(a,b)(2-\psi)^{a+b}}$$

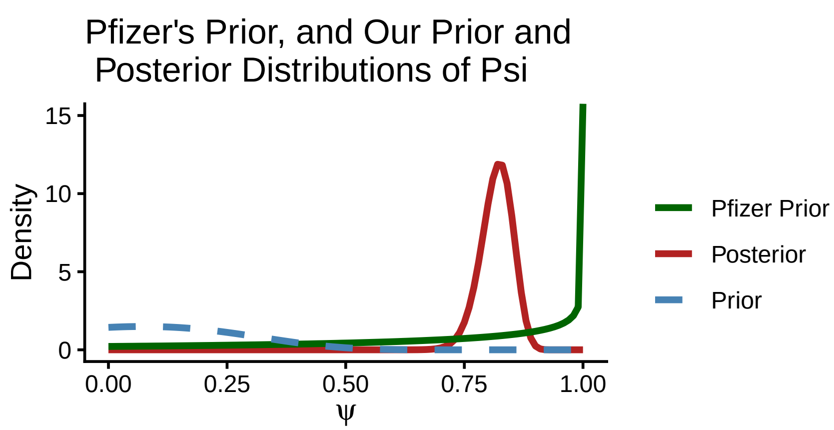
Substituting the $\text{Beta}(34.2, 188.2)$ density gives

$$f_{\Psi|V}(\psi \mid v) = \frac{(1-\psi)^{33.2}}{B(34.2, 188.2)(2-\psi)^{222.4}}, \qquad -\infty < \psi \le 1.$$

Similarly, the prior density of Ψ is

$$f_{\Psi}(\psi) = \frac{(1-\psi)^{25.2}}{B(26.2, 26.2)(2-\psi)^{52.4}}, \qquad -\infty < \psi \le 1.$$

The prior and posterior densities (as well as the Pfizer study's prior distribution for comparison) for ψ are shown below.



Pfizer prior, the elicited prior, and the posterior distribution for the vaccine efficacy

Minding the asymptote, the plot suggests that Pfizer's prior of $\Pi \sim B(0.700102, 1)$ is optimistic and assumes the vaccine is efficacious before the outcsome of the trial.

Posterior summaries of Ψ can also be obtained from the posterior distribution of Π using the transformation $\Psi = \frac{1-2\Pi}{1-\Pi}$.

The posterior median of Ψ is defined by the value $q_{\Psi,0.5}$ satisfying

$$P(\Psi \leq q_{0.5} \mid V = v_{obs}) = 0.5.$$

Using the transformation between Ψ and Π,

$$\begin{aligned}
P(\Psi \leq q_{\Psi|V,0.5} \mid V = v_{obs}) &= P\left(\frac{1-2\Pi}{1-\Pi} \leq q_{\Psi|V,0.5} \,\middle|\, V = v_{obs}\right) \\
&= P\left(\Pi(q_{\Psi|V,0.5} - 2) \leq q_{\Psi|V,0.5} - 1 \,\middle|\, V = v_{obs}\right) \\
&= P\left(\Pi \geq \frac{q_{\Psi|V,0.5} - 1}{q_{\Psi|V,0.5} - 2} \,\middle|\, V = v_{obs}\right) = 0.5
\end{aligned}$$

$$P\left(\Pi \leq \frac{q_{\Psi|V,0.5} - 1}{q_{\Psi|V,0.5} - 2} \,\middle|\, V = v_{obs}\right) = 1 - 0.5$$

A similar transformation can be used to obtain the 95% credible interval for Ψ. Let $q_{\Psi|V,0.025}$ and $q_{\Psi|V,0.975}$ denote the 2.5th and 97.5th percentiles of the posterior distribution of Ψ. The corresponding endpoints are

$$q_{\Psi|V,0.025} = \frac{1 - 2q_{\Pi|V,0.975}}{1 - q_{\Pi|V,0.975}}, \qquad q_{\Psi|V,0.5} = \frac{1 - 2q_{\Pi|V,0.5}}{1 - q_{\Pi|V,0.5}} \qquad q_{\Psi|V,0.975} = \frac{1 - 2q_{\Pi|V,0.025}}{1 - q_{\Pi|V,0.025}}$$

To assess whether the vaccine meets the FDA efficacy guideline, we compute the posterior probability

$$P(\Psi \leq 0.3 \mid V = v),$$

which can be evaluated either directly from the posterior density of Ψ or by transforming the equivalent event on Π.

Results

Likelihood Inference Results

Utilizing the log likelihood functions and its derivatives, we identify a critical point and verify it to be the MLE. We find it to be equivalent to the estimate we computed at the beginning using the transformation of the binomial MLE. That is, $\widehat{\psi}_0^{mle} \approx 0.9506$.

With the observed Fisher Information, we find our large sample confidence interval to be [0.9155, 0.9857]. On the other hand, the parametric bootstrap with 100000 samples returns a confidence interval of [0.9102564, 0.9820359].

The significance test returns $2.8223132 \times 10^{-28}$, which is comparable to our empirical p-value, 10^{-5}. Since none of the bootstrapped values were greater than our observed likelihood ratio statistic, W, our empirical P-value was calculated as $\frac{1}{\text{bootstrap samples}+1}$.

Bayesian Inference Results

Using the posterior distribution of $\Pi \mid V = v \sim \text{Beta}(34.2, 188.2)$ and the transformation discussed in the Method,

$$q_{\Psi|V,0.025} = \frac{1 - 2q_{\Pi|V,0.975}}{1 - q_{\Pi|V,0.975}}, \qquad q_{\Psi|V,0.5} = \frac{1 - 2q_{\Pi|V,0.5}}{1 - q_{\Pi|V,0.5}} \qquad q_{\Psi|V,0.975} = \frac{1 - 2q_{\Pi|V,0.025}}{1 - q_{\Pi|V,0.025}}$$

we obtained a posterior median of 0.82, and a 95% credible interval of (0.744, 0.877).

We can also use the posterior density of Ψ derived in the previous section and numerical integration to compute summary statistics for the posterior distribution of the vaccine efficacy parameter ψ. In particular, the posterior mean, posterior median, and a 95% Bayesian credible interval were obtained.

Table 2: Posterior summary statistics for Psi.

Mean	Median	Lower.Bound	Upper.Bound
0.817	0.82	0.744	0.877

Calculate a Bayesian P-value

To assess whether the vaccine meets the FDA efficacy guideline, we consider the hypothesis test

$$H_0 : \psi = 0.3 \quad H_1 : \psi > 0.3$$

Using the posterior distribution of ψ, we compute the posterior probability

$$P(\Psi \leq 0.3 \mid V = v),$$

which serves as the Bayesian p-value for this one-sided test.

The posterior probability is

$$P(\Psi \leq 0.3 | V = v) = 2.959 \cdot 10^{-17}$$

Since the posterior probability that $\psi \leq 0.3$ is essentially zero, we can reject the null hypothesis.

Discussion / Conclusion

Table 3: '95% CI for Psi'

	Lower Bound	Upper Bound
MLE Confidence Interval	0.9155	0.9857
Our Credible Interval	0.7439	0.8770
Pfizer's Credible Interval	0.9030	0.9760

Both our frequentist and Bayesian analysis corroborate Pfizer's finding that the BNT162b2 vaccine was 95% effective in preventing COVID-19. Similar to Pfizer's original report, our significance tests for both Bayesian and frequentist estimates of ψ were very statistically significant, with a p-value well below 0.01 or any other reasonable rejection level. This means we should reject the null hypothesis that the vaccine efficacy does not exceed the FDA guidelines and it is highly probable that the vaccine efficacy is above FDA minimum guidelines. Our empirical tests for both Bayesian and frequentist methods also support the rejection of the null hypothesis.

While all three tests validate that the BNT162b2 vaccine was more efficacious than FDA guidelines ($\psi > 0.3$) at a 95% CI, our results show that Pfizer overestimated the efficacy of their vaccine based on Bayesian methods. The confidence interval for ψ^{mle} is similar to Pfizer's credible interval for ψ, our posterior median's lower bound for our median was ~15% smaller and our posterior median's upper bound was ~10% smaller than Pfizer's credible interval bounds. This overestimation mainly comes from the difference in choosing the prior. While Pfizer themselves state their prior of $\psi = 0.3$ "can be considered pessimistic", this is far from the case. This prior makes the implicit assumption that the vaccine is more efficacious before seeing any trial (BioNTech/Pfizer, 2020). They assumed π is centered at 0.4118, which is lower than 0.5. A prior π of 0.5 assumes that the efficacy of a placebo and the vaccine are the same. Our prior does not encode the assumption that the drug will be efficacious, as the most rigorous standard for clinical trials is to assume no efficacy from any given medical intervention.

In the future, large pharmaceutical companies that use Bayesian methods should be more rigorous and transparent about their selection of a prior. We have demonstrated that through selecting a more rigorous prior, prior selection can impact the magnitude of final estimates significantly- even if the treatment is highly effective. Without fully transparent methodology, fellow researchers and the general public can become skeptical of one's findings which can create material harms in public health outreach.

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Appendix

Code

Setup

```
#Use this code chunk to include libraries, and set global options.
library(ggplot2)
library(tidyverse)
library(fastR2)
```

Data visualization

```
df <- data.frame(
  Group = c('BNT162b2', 'Placebo'),
  cases = c(8, 162),
  N = c(17411, 17511),
  rate = c(8/17411, 162/17511)
)
knitr::kable(df, digits = 5, caption =  "vaccine efficacy at least 7 days after dose two"
)
ggplot(df, aes(x = Group, y = rate, fill = Group))+
  geom_col()+ scale_fill_manual(
    values = c('BNT162b2' = 'steelblue', 'Placebo' = 'firebrick')
  )+labs(x = "Group",y = "Infection Rate",title = "Observed Infection Rates by Group",
         caption = "Figure 1: Observed Infection Rates by Group") +
  theme_classic()+ theme(legend.position = "none", plot.caption = element_text(hjust = 0.5))
```

Plot second-order taylor approximation against log-likelihood

```
loglik_pi <- function(pi, n, v){
  c <- choose(n, v)
  return(log(c) + (v * log(pi)) + ((n-v) * log(1 - pi)))
}
ml_pi <- maxLik2(loglik = loglik_pi,
                    start = 0.5, n = 170, v = 8)
plot(ml_pi) +
  labs(title = "Second order approximation to the pi likelihood",
  x = expression(pi),
  caption = "Figure 2: Log-likelihood function and second-order approximation for pi."
  )+
  theme_classic()+
  theme(plot.caption = element_text(hjust = 0.5))
```

Compute large sample confidence interval for the MLE

```
z <- qnorm(0.975)
I <- (8/(1 - 0.9506)^2) - (170/(2-0.9506)^2)
upper <- 0.9506 + z * (1/sqrt(I))
lower <- 0.9506 - z * (1/sqrt(I))
```

Compute pvalue using likelihood ratio statistic

```
pval <- pchisq(121.6012, df = 1, lower.tail = F)
```

Parametric Bootstrap for psi

```
set.seed(342)

W.func <- function(n, v, mle, null) {
  W <- 2*(v*log(1-mle) + n*log(2-null) - n*log(2-mle) - v*log(1-null))
  return(W)
}

psi_mle <- (1 - 2*ml_pi$estimate)/(1-ml_pi$estimate)
W <- W.func(170, 8, psi_mle, 0.3)

B <- 100000

big <- replicate(B, W.func(n = 170, v = rbinom(1, 170, prob = 0.7/1.7),
                mle = psi_mle, null = 0.3))

sum(big > W)

## Simulate values of psi

psi.func <- function(n, obs) {
 view <- rbinom(1, n, prob = obs)/n
  psi <- (1- 2*view)/(1-view)
  return(psi)
}

psi.sim <- replicate(B, psi.func(n = 170, obs = 8/170))
lower.sim <- quantile(psi.sim, c(0.025, 0.975))[[1]]
upper.sim <- quantile(psi.sim, c(0.025, 0.975))[[2]]
```

Find Prior

```
library(LearnBayes)
beta.select(quantile1 = list(p = 0.1, x = 7/17),
            quantile2 = list(p = 0.5, x = 0.5))
```

Compute statistics for psi

```
post_psi <- function(x) {(1-x)^33.2/(beta(34.2, 188.2)*(2-x)^222.4)}
exp_post_psi<- function(x) {x*(1-x)^33.2/(beta(34.2, 188.2)*(2-x)^222.4)}
find_quantile <- function(p){
  uniroot(f = function(q) {
    integrate(post_psi, 0, q)$value - p},
    interval = c(0,1))$root}
q_025 <- find_quantile(0.025)
q_median <- find_quantile(0.5)
q_975<- find_quantile(0.975)
psi_mean <- integrate(exp_post_psi, 0, 1)$value
result_table <- data.frame(Mean = psi_mean, Median = q_median,
                           'Lower Bound' = q_025, 'Upper Bound' = q_975)
knitr::kable(result_table, digits = 3, caption = "Posterior summary statistics for Psi.")
```

Plot of posterior distribution of psi

```
prior_psi <- function(x) {(1-x)^25.2/(beta(26.2,26.2)*(2-x)^52.4)}
post_psi <- function(x) {(1-x)^33.2/(beta(34.2, 188.2)*(2-x)^222.4)}
pfizer_prior <- function(x) {(1-x)^(-0.299898) / (beta(0.700102, 1) * (2-x)^1.700102)}
ggplot() +
geom_function(fun = post_psi, xlim = c(0,1) ,aes(color = "Posterior"), linewidth = 1.2) +
geom_function(fun = pfizer_prior, xlim = c(0,1), aes(color = "Pfizer Prior"), linewidth = 1.2)
geom_function(fun = prior_psi, xlim = c(0,1),
              aes(color = "Prior"), linewidth = 1.2, linetype = 'dashed') +
scale_color_manual(values = c("Posterior" = "firebrick","Prior" = "steelblue", "Pfizer Prior" =
labs(x = expression(psi), y = "Density", color = "",
     title = "Pfizer's Prior, and Our Prior and\n Posterior Distributions of Psi",
    caption = "Comparison of the Pfizer prior, the elicited prior, and the posterior distributi
theme_classic()+
  coord_cartesian(ylim = c(0, 15))+
  theme(plot.caption = element_text(hjust = 0.5))
```

Computing 50th, 2.5th, 0.975th of Psi

```
inverse <- qbeta(0.5, 34.2, 188.2)
q_05 <- (2*inverse - 1)/(inverse - 1 )
inverse_025 <- qbeta(0.025, 34.2, 188.2)
inverse_975 <- qbeta(0.975, 34.2, 188.2)
q_025 <- (1- 2*inverse_975)/(1-inverse_975)
q_975 <- (1- 2*inverse_025)/(1-inverse_025)
```

Parametric Boostrap

```r
set.seed(342)

W.func <- function(n, v, mle, null) {
  W <- 2*(v*log(1-mle) + n*log(2-null) - n*log(2-mle) - v*log(1-null))
  return(W)
}

psi_mle <- (1 - 2*ml_pi$estimate)/(1-ml_pi$estimate)
W <- W.func(170, 8, psi_mle, 0.3)

B <- 100000

big <- replicate(B, W.func(n = 170, v = rbinom(1, 170, prob = 0.7/1.7),
                mle = psi_mle, null = 0.3))

sum(big > W)

## Simulate values of psi

psi.func <- function(n, obs) {
 view <- rbinom(1, n, prob = obs)/n
  psi <- (1- 2*view)/(1-view)
  return(psi)
}

psi.sim <- replicate(B, psi.func(n = 170, obs = 8/170))
lower.sim <- quantile(psi.sim, c(0.025, 0.975))[[1]]
upper.sim <- quantile(psi.sim, c(0.025, 0.975))[[2]]
```

Table Code

```r
library(knitr)
library(latex2exp)
ci.df <- data.frame(Lower_Bound = c(lower, q_025, 0.903),
Upper_Bound = c(upper, q_975, 0.976))
rownames(ci.df) <- c("MLE Confidence Interval" ,
"Our Credible Interval", "Pfizer's Credible Interval")
colnames(ci.df) <- c("Lower Bound", "Upper Bound")

kable(ci.df,
      digits = 4,
      caption = TeX("95% CI for Psi", output = "character"))
```

Calculate Bayesian p

```r
integrate(post_psi, 0, 0.3)
```

Proofs

Proving that the random variable `V`, representing the proportion of people who were infected with COVID that were in the vaccine group, approximates to the binomial distribution.

Group	Cases	Sample size
BNT162b2	8	17,411
Placebo	162	17,511
Total	170	34,922

From the above data, we define X as a random variable representing the number of subjects in the vaccine group (of size n_1) who were infected with COVID, and Y as a random variable representing the number of subjects in the placebo group (of size n_2) who were infected with COVID. We assume they take the binomial distribution as follows:

$$X \sim Binom(n_1, \pi_v) \; Y \sim Binom(n_2, \pi_p)$$

Where $\pi_v = P(\text{COVID}|\text{Vaccine})$, $\pi_p = P(\text{COVID}|\text{Placebo})$, and X and Y are independent.
We then define V as a random variable that represents the number of subjects that were infected COVID that were in the vaccine group. Then, V is the value of X given the constraint that $X + Y = n$, where n is the sample size of our variable V. (which, in our data, is $n = 170$). We define π as the rate at which a subject in the sample size n is in the vaccine group.
As the sample sizes n_1 and n_2 are quite large, we know that the binomial random variables X and Y can be approximated with the Poisson. That is,

$$X \approx Pois(n_1\pi_v) \; Y \approx Pois(n_2\pi_p)$$

Where $\lambda_x = E[X] = n_1\pi_v$, and similarly for $\lambda_y = E[Y]$
Before we proceed with the proof, we will prove two Lemmas, which will help with our main argument.

[**Lemma I**] Theorem: The rate of a COVID patient being in the vaccine group, π, is the probability $P(\text{Vaccine}|\text{COVID})$. Then:

$$\pi = \frac{P(\text{COVID}|\text{Vaccine}) \cdot P(\text{Vaccine})}{P(\text{COVID})} = \frac{n_1\pi_v}{\pi_v n_1 + \pi_p n_2}$$

Proof:
Using the data and the distribution definitions above:

$$\begin{aligned} P(\text{COVID}|\text{Vaccine}) &= \pi_v \\ P(\text{Vaccine}) &= \frac{n_1}{n_1 + n_2} \\ P(\text{COVID}) &= \frac{\pi_v n_1 + \pi_p n_2}{n_1 + n_2} \end{aligned}$$

Thus,

$$\pi = \frac{\pi_v \cdot \frac{n_1}{n_1+n_2}}{\frac{\pi_v n_1 + \pi_p n_2}{n_1+n_2}}$$
$$\pi = \frac{n_1 \pi_v}{\pi_v n_1 + \pi_p n_2}\square$$

[Lemma II] Theorem: Let X and Y be two independent Poisson random variables. Then, the sum of X and Y is also a Poisson distribution. That is,

$$S \sim Pois(\lambda_1 + \lambda_2) \quad S = X + Y$$

Proof:
Let $X \sim Pois(\lambda_1)$ and, independently, $Y \sim Pois(\lambda_2)$. Suppose $S = X + Y$
Then:

$$P(X = x) = \frac{e^{-\lambda_1}\lambda_1^x}{x!}$$
$$P(Y = n - x) = \frac{e^{-\lambda_2}\lambda_2^{n-x}}{(n-x)!}$$

By theorem 3.1 (Grove, 2025, Lecture 3):

$$P(S = n) = \sum_{x=0}^{n} P(X = x)P(Y = n - x)$$
$$P(S = n) = \sum_{x=0}^{n} \frac{e^{-\lambda_1}\lambda_1^x}{x!} \cdot \frac{e^{-\lambda_2}\lambda_2^{n-x}}{(n-x)!}$$
$$P(S = n) = e^{-\lambda_1+\lambda_2} \sum_{x=0}^{n} \frac{\lambda_1^x \lambda_2^{n-x}}{x!(n-x)!}$$

$\sum_{x=0}^{n} \frac{\lambda_1^x \lambda_2^{n-x}}{x!(n-x)!}$ is the binomial expansion of $\frac{(\lambda_1+\lambda_2)^n}{n!}$. So,

$$P(S = n) = \frac{e^{-\lambda_1+\lambda_2}(\lambda_1 + \lambda_2)^n}{n!}$$

Thus, from the PMF above:

$$X + Y = S \sim Pois(\lambda_1 + \lambda_2)\square$$

Main proof
Theorem: The random variable V, defined above, approximates to the binomial distribution.
Proof:
We proceed by finding the Probability Mass Function (PMF) of V. We know V to be the value of X given the constraint that $X + Y = n$. Our process, minding the independence of X and Y, thus follows:

$$P(V=v) = P(X=v|X+Y=n)$$
$$P(V=v) = \frac{P(X=v \cap X+Y=n)}{P(X+Y=n)}$$
$$P(V=v) = \frac{P(X=v) \times P(Y=n-v)}{P(X+Y=n)}$$

As X and Y can be approximated with the Poisson distribution, we will use the Poisson PMF, $\frac{e^{-\lambda}\lambda^x}{x!}$.
Remember that $X \approx Pois(n_1\pi_v)$ and $Y \approx Pois(n_2\pi_p)$. And so, by Lemma II, $X+Y \approx Pois(n_1\pi_v + n_2\pi_p)$

$$P(V=v) = \frac{\frac{e^{-(n_1\pi_v)}(n_1\pi_v)^v}{v!} \times \frac{e^{-(n_2\pi_p)}(n_2\pi_p)^{n-v}}{(n-v)!}}{\frac{e^{-(n_1\pi_v+n_2\pi_p)}(n_1\pi_v+n_2\pi_p)^n}{n!}}$$
$$P(V=v) = \frac{n!}{v!(n-v)!} \times \frac{(n_1\pi_v)^v(n_2\pi_p)^{n-v}e^{-(n_1\pi_v+n_2\pi_p)}}{(n_1\pi_v+n_2\pi_p)^n e^{-(n_1\pi_v+n_2\pi_p)}}$$
$$P(V=v) = \binom{n}{v}\frac{(n_1\pi_v)^n(n_2\pi_p)^{n-v}}{(n_1\pi_v+n_2\pi_p)^v(n_1\pi_v+n_2\pi_p)^{n-v}}$$

We apply Lemma I, to simplify the following:

$$\frac{(n_1\pi_v)^v}{(n_1\pi_v+n_2\pi_p)^v} = (\frac{n_1\pi_v}{\pi_v n_1 + \pi_p n_2})^v = \pi^v$$
$$(\frac{n_2\pi_p}{n_1\pi_v+n_2\pi_p})^{n-v} = (1 - \frac{n_1\pi_v}{\pi_v n_1 + \pi_p n_2})^{n-v} = (1-\pi)^{n-v}$$

Thus,

$$P(V=v) = \binom{n}{v}\pi^v(1-\pi)^{n-v}$$

The PMF of the binomial distribution. Therefore:

$$V \approx Binom(n,\pi) \blacksquare$$